Stable Maps: Basic Definitions

YunKai CHI

This lecture introduces pointed nodal curves, stable maps, and their coarse moduli spaces. Stability is expressed both by positivity of a line bundle and by finiteness of automorphisms. The lecture concludes with convex targets, the genus-zero boundary, and the construction of evaluation and forgetful maps.

Contents

1 Stable Maps and Their Moduli Spaces **1**
1.1 Pointed Curves and Stable Maps 1
1.1A Quasi-stable curves 1
1.1B Families of maps and their isomorphisms 2
1.1C Special points and the stability condition 2
1.2 Stability as Relative Ampleness 3
1.2A The relative dualizing sheaf and the markings 3
1.2B Degree on each irreducible component 4
1.3 Curve Classes and the Moduli Functor 5
1.3A The class represented by a map 5
1.3B The set-valued functor 5
1.4 Existence and the Coarse Universal Property 6
1.5 Stability and Finiteness of Automorphisms 7
1.5A The automorphism-free locus 10
1.6 Convex Targets and the Genus-Zero Boundary 10
1.6A Reducible source curves 10
1.6B The target is a point 11
1.7 Evaluation Maps and Forgetting the Map 12
1.7A Independence of a projective embedding 12
1.7B Evaluation as a natural transformation 12
1.7C Forgetting the map and stabilizing the source 13

References **14**

Chapter 1

Stable Maps and Their Moduli Spaces

We work over $\mathbb{C}$, as in [FP97]. The definitions of pointed curves and stability also make sense over an algebraically closed field of characteristic zero. Singular homology and the notation X^{an} will be used only over $\mathbb{C}$. Test schemes may be nonreduced, and families are required to be of finite presentation; for the sheaf-theoretic arguments, bases are locally noetherian.

The principal reference is [FP97], §1, *Stable maps and their moduli spaces*. We use the January 5, 2026 draft of Alper's *Stacks and Moduli* [Alper26]. Throughout the lecture, $g, n \geq 0$ are integers. Markings are ordered and must be preserved individually by isomorphisms.

1.1 Pointed Curves and Stable Maps

1.1A Quasi-stable curves

Definition 1.1.1 (Quasi-stable curve)**.** An *n-pointed, arithmetic-genus-g, quasi-stable curve*

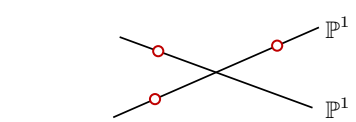
$$(C; p_1, \ldots, p_n)$$

is a projective, connected, reduced curve of arithmetic genus g, with at worst nodes, together with n distinct nonsingular marked points. Here

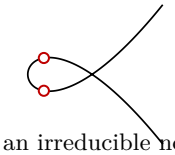
$$p_a(C) = 1 - \chi(C, \mathcal{O}_C) = g.$$

The term *quasi-stable* follows [FP97]; it is often called *prestable*. No stability inequality is imposed in Definition 1.1.1. In particular, this use of "quasi-stable" should not be confused with other conventions which impose additional conditions on rational components.

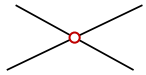
Example 1.1.2 (Allowed and excluded marked curves)**.** Nodes are allowed, but a marked point may not be a node. A cusp is not an allowed singularity. In the following schematic drawings, red circles are markings; a crossing represents a node. Components drawn as arcs or lines are complete curves, not affine pieces of the domain.



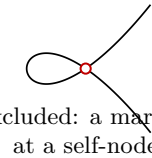
Allowed: markings away from the node



Allowed: an irreducible nodal curve



Excluded: a marking at a node



Excluded: a marking at a self-node



Excluded: a cusp

Definition 1.1.3 (Family of quasi-stable curves)**.** Let S be a scheme over $\mathbb{C}$. A *family of n-pointed, arithmetic-genus-g, quasi-stable curves* over S is a flat projective morphism of finite presentation

$$\pi : \mathcal{C} \longrightarrow S$$

with sections $p_1, \dots, p_n : S \to \mathcal{C}$ such that, for every geometric point $\bar{s} \to S$,

$$(\mathcal{C}_{\bar{s}}; p_1(\bar{s}), \dots, p_n(\bar{s}))$$

is a curve as in Definition 1.1.1. Here $\mathcal{C}_{\bar{s}} = \mathcal{C} \times_S \operatorname{Spec} \overline{k(s)}$, and the markings on this fibre are the base changes of the sections.

Over a noetherian base, finite presentation follows from finite type; it is written explicitly to permit more general test schemes. The fibres are geometrically reduced and connected. For such a proper flat family, the natural map

$$\mathcal{O}_S \xrightarrow{\sim} \pi_* \mathcal{O}_{\mathcal{C}}$$

is an isomorphism, and the same holds after base change. See [Stacks], Tag 0E0L.

1.1B Families of maps and their isomorphisms

Definition 1.1.4 (Family of pointed maps)**.** Let X be a scheme over $\mathbb{C}$. A *family of maps to X* over S consists of data

$$\xi = (\pi : \mathcal{C} \to S, \{p_i\}_{i=1}^n, \mu : \mathcal{C} \to X),$$

where $(\pi, \{p_i\})$ is a family of quasi-stable curves and μ is a morphism. The curve varies with S; the target X is fixed.

Two such families $\xi = (\pi, \{p_i\}, \mu)$ and $\xi' = (\pi', \{p_i'\}, \mu')$ over S are *isomorphic* if there is a scheme isomorphism $\tau : \mathcal{C} \xrightarrow{\sim} \mathcal{C}'$ with

$$\pi = \pi' \circ \tau, \qquad p_i' = \tau \circ p_i, \qquad \mu = \mu' \circ \tau.$$

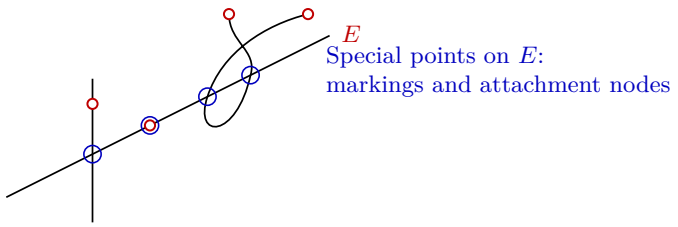
When $S = \operatorname{Spec} \mathbb{C}$, we write $(C; \{p_i\}, \mu)$.

Thus an isomorphism is allowed to reparameterize the domain, but it must preserve the ordered markings and the map to X.

1.1C Special points and the stability condition

Definition 1.1.5 (Special points on a component)**.** Let $E \subset C$ be an irreducible component. Following the convention of [FP97], its *special points* are the marked points on E and the nodes where E meets *other* irreducible components. Let s_E be their number. Self-nodes of E are not included in s_E; they are already reflected in $p_a(E)$.

When working on the normalization $\widetilde{E} \to E$, one instead counts the inverse images of *all* nodes together with the markings. A self-node then contributes two distinct points. This distinction will be used explicitly in the proof of the automorphism criterion.



Definition 1.1.6 (Stable map)**.** A pointed map $(C; \{p_i\}, \mu)$ is *stable* if every irreducible component $E \subset C$ satisfies the following conditions:

(1) If $E \simeq \mathbb{P}^1$ and μ contracts E to a point, then $s_E \geq 3$.

(2) If $p_a(E) = 1$ and μ contracts E to a point, then $s_E \geq 1$.

A family of pointed maps is stable if its restriction to every geometric fibre is stable.

There is no additional marking requirement on a noncontracted component. There is also no additional condition on a contracted component of arithmetic genus at least two. For a smooth contracted rational component, three fixed special points remove the continuous freedom of projective reparameterization. For a smooth contracted elliptic component, one fixed point removes translations.

References. Definitions 1.1.1–1.1.6 follow [FP97], §1.1, pp. 10–11. For pointed stable curves and normalization conventions, compare [Alper26], Definitions 6.3.2 and 6.3.9, pp. 225 and 227.

1.2 Stability as Relative Ampleness

1.2A The relative dualizing sheaf and the markings

Let the target be $\mathbb{P}^r$, and let $(\pi : \mathcal{C} \to S, \{p_i\}, \mu)$ be a family of pointed maps. The relative dualizing sheaf $\omega_{\mathcal{C}/S}$ is a line bundle, and on a geometric fibre $C = \mathcal{C}_{\bar{s}}$ it restricts to ω_C.

Indeed, a nodal curve is locally a complete intersection, hence Gorenstein and Cohen–Macaulay. Relative duality for a flat family of curves gives a relative dualizing complex of the form

$$\omega^{\bullet}_{\mathcal{C}/S} = \omega_{\mathcal{C}/S}[1],$$

and the Gorenstein condition makes $\omega_{\mathcal{C}/S}$ invertible and compatible with base change. These assertions are recorded in [Stacks], Tag 0E6P and Tag 0E6R.

The images $p_i(S)$ are effective Cartier divisors. To see this, the fibrewise condition says that $p_i(\bar{s})$ lies in the smooth locus of every geometric fibre. Flatness and finite presentation imply that π is smooth along each section. Since π is separated, each section is a closed immersion; a section of a smooth morphism of relative dimension one is an effective Cartier divisor. Consequently the line bundle

$$\mathcal{L} = \omega_{\mathcal{C}/S}\Big(\sum_{i=1}^{n} p_i(S)\Big) \otimes \mu^{*}\mathcal{O}_{\mathbb{P}^r}(3) \tag{1.2.1}$$

is well defined.

References. Smoothness along sections: [Stacks], Tag 01V4; regular immersions given by sections: Tag 067R. For the dualizing line of a nodal curve, see [Alper26], Definition 6.2.18, p. 220; for its pullback to normalized components, see the proof of Proposition 6.3.5, p. 226.

1.2B Degree on each irreducible component

Fix a geometric fibre C and an irreducible component $E \subset C$. Write

$$b_E = \#\{\text{nodes joining } E \text{ to other components}\},$$
$$m_E = \#\{\text{markings on } E\}, \qquad s_E = b_E + m_E.$$

Let $q_1, \dots, q_{b_E}$ be those attachment nodes. The restriction formula is

$$\omega_C|_E \simeq \omega_E(q_1 + \cdots + q_{b_E}). \tag{1.2.2}$$

Self-nodes are included in the intrinsic dualizing sheaf ω_E. Formula (1.2.2) therefore holds also when E itself is singular.

For a line bundle A on the proper integral curve E, its degree is

$$\deg_E A = \chi(E, A) - \chi(E, \mathcal{O}_E).$$

Serre duality gives $\chi(E, \omega_E) = -\chi(E, \mathcal{O}_E)$, so

$$\begin{aligned} \deg_E(\omega_C|_E) &= \chi(E, \omega_E) + b_E - \chi(E, \mathcal{O}_E) \\ &= -2\chi(E, \mathcal{O}_E) + b_E \\ &= 2p_a(E) - 2 + b_E. \end{aligned}$$

Put $d_E = \deg_E \mu^* \mathcal{O}_{\mathbb{P}^r}(1)$. It is zero if E is contracted, and is a positive integer otherwise. Thus

$$\deg_E(\mathcal{L}|_C) = 2p_a(E) - 2 + s_E + 3d_E. \tag{1.2.3}$$

Proposition 1.2.1 (The ampleness criterion)**.** *A family of pointed maps to $\mathbb{P}^r$ is stable if and only if the line bundle* (1.2.1) *is relatively ample over S.*

Proof. Relative ampleness for this proper family can be checked on its geometric fibres. On a proper curve a line bundle is ample precisely when it has positive degree on every irreducible component. Apply (1.2.3).

If $\mathcal{L}$ is relatively ample and E is contracted with $p_a(E) = 0$, positivity gives $-2+s_E > 0$, hence $s_E \geq 3$. If $p_a(E) = 1$, it gives $s_E > 0$, hence $s_E \geq 1$. These are the stability conditions.

Conversely, suppose the family is stable. On a contracted component the degree is positive by those conditions, or automatically if $p_a(E) \geq 2$. On a noncontracted component, $d_E \geq 1$ and

$$2p_a(E) - 2 + s_E + 3d_E \geq -2 + 3 = 1.$$

Every component has positive degree, so $\mathcal{L}$ is relatively ample. □

The exponent three makes the positivity on noncontracted rational components automatic, even when they have no special points.

References. The criterion is in [FP97], §1.1, p. 11. Degree and componentwise ampleness: [Stacks], §33.44, especially Tag 0B5Y. For the analogous family criterion for stable pointed curves, see [Alper26], Proposition 6.3.22, p. 231.

1.3 Curve Classes and the Moduli Functor

1.3A The class represented by a map

Definition 1.3.1 (A map representing a class)**.** Let X be a scheme of finite type over $\mathbb{C}$ and $\beta \in H_2(X^{\mathrm{an}}, \mathbb{Z})$. A map $\mu : C \to X$ from a quasi-stable curve *represents* β if

$$\mu_*[C] = \beta$$

in singular homology. For a family over a finite-type complex base, the fixed-class condition means that every complex fibre map represents β. This is the usual class condition in the moduli functor below; on more general test schemes it is obtained by descent from finite-type bases.

The class of the domain is the sum of its component classes. A noncontracted component contributes the class of its image with the degree of the map as multiplicity; a contracted component contributes zero.

Remark 1.3.2 (The corresponding algebraic one-cycle)**.** If X is separated, every map $\mu : C \to X$ from a projective curve is proper. Indeed, the graph gives the factorization

$$\mu : C \xrightarrow{\Gamma_\mu} C \times_{\mathbb{C}} X \xrightarrow{\mathrm{pr}_2} X, \qquad c \longmapsto (c, \mu(c)) \longmapsto \mu(c).$$

The graph is a closed immersion, and the projection is proper. We can therefore also push forward $[C]$ in the Chow group. This gives an effective one-cycle, with multiplicities recording covering degrees.

For smooth projective homogeneous varieties, including projective spaces and Grassmannians, the cycle-class map identifies the Chow groups with the corresponding even homology groups. In that setting one may write

$$\mathrm{CH}_d(X) \simeq H_{2d}(X^{\mathrm{an}}, \mathbb{Z}).$$

The word *projective* is relevant here; this identification is not being asserted for arbitrary nonproper homogeneous varieties.

References. See [FP97], §0.2 and §1.1. For algebraic cycles, proper pushforward, and the cycle-class map, see [Fulton98], Chapter 1 and §19.1.

1.3B The set-valued functor

Definition 1.3.3 (The stable-maps functor)**.** Define a contravariant functor

$$\overline{\mathcal{M}}_{g,n}(X, \beta) : (\mathrm{Sch}/\mathbb{C})^{\mathrm{op}} \longrightarrow \mathrm{Sets}$$

by letting $\overline{\mathcal{M}}_{g,n}(X, \beta)(S)$ be the set of isomorphism classes of stable families of maps to X over S, with n markings, arithmetic genus g, and class β. For a morphism $a : T \to S$, the induced map sends a family to its pullback:

$$(\pi : \mathcal{C} \to S, \{p_i\}, \mu) \longmapsto (\pi_T : \mathcal{C} \times_S T \to T, \{p_{i,T}\}, \mu_T).$$

This is contravariant because a map of test schemes $T \to S$ reverses the direction of the map on families. The notation is that of [FP97]: in this lecture $\overline{\mathcal{M}}_{g,n}$ denotes the *set-valued isomorphism-class functor*. A modern moduli stack instead retains the groupoids of families and their automorphisms. These two objects should not be identified.

Remark 1.3.4 (Smooth sources and nodal sources)**.** The smooth-source locus $M_{g,n}(X,\beta)$ consists of stable pointed maps $(C;\{p_i\},\mu)$ with C nonsingular of genus g. The same isomorphism condition applies: an isomorphism of curves must preserve each marking and commute with the map to X.

When $\beta \neq 0$, the space is empty unless β is represented by an effective curve cycle. In genus zero it is empty if X contains no rational curves. Allowing connected nodal sources gives the proper moduli space $\overline{M}_{g,n}(X,\beta)$ constructed below, containing the smooth-source locus as an open subset. For a general target, the word "compactification" should not be taken to assert that this open subset is dense in every component.

If $X = \operatorname{Spec}\mathbb{C}$ and $\beta = 0$, stable maps are just stable pointed curves, so

$$\overline{M}_{g,n} \simeq \overline{M}_{g,n}(\operatorname{Spec}\mathbb{C}, 0).$$

Some older references, including [Knudsen83], use notation without the bar for the space of stable nodal curves. Here the bar distinguishes it from the smooth-curve locus.

1.4 Existence and the Coarse Universal Property

Theorem 1.4.1 (Existence of the coarse moduli space)**.** *If X is projective over $\mathbb{C}$, there exists a projective coarse moduli scheme*

$$\overline{M}_{g,n}(X,\beta).$$

This is [FP97], §1.2, Theorem 1. Its proof is developed in §§2–5 of that reference. We explain the universal property, rather than reproduce the construction.

Write $h_Z(S) = \operatorname{Hom}_{\mathbb{C}}(S,Z)$ for the functor of points of a $\mathbb{C}$-scheme Z. The theorem supplies a natural transformation

$$\varphi : \overline{\mathcal{M}}_{g,n}(X,\beta) \longrightarrow h_{\overline{M}_{g,n}(X,\beta)} \tag{1.4.1}$$

with the following properties:

(I) On $\mathbb{C}$-points, the map

$$\varphi_{\operatorname{Spec}\mathbb{C}} : \overline{\mathcal{M}}_{g,n}(X,\beta)(\operatorname{Spec}\mathbb{C}) \longrightarrow \operatorname{Hom}_{\mathbb{C}}(\operatorname{Spec}\mathbb{C}, \overline{M}_{g,n}(X,\beta))$$

is a bijection.

(II) For any $\mathbb{C}$-scheme Z and natural transformation

$$\psi : \overline{\mathcal{M}}_{g,n}(X,\beta) \longrightarrow h_Z,$$

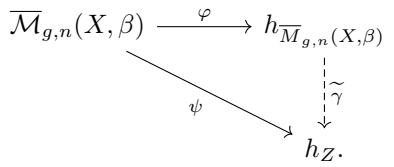
there is a unique morphism

$$\gamma : \overline{M}_{g,n}(X,\beta) \longrightarrow Z$$

such that $\psi = \widetilde{\gamma} \circ \varphi$. Here

$$\widetilde{\gamma}_S : \operatorname{Hom}_{\mathbb{C}}(S, \overline{M}_{g,n}(X,\beta)) \to \operatorname{Hom}_{\mathbb{C}}(S,Z), \quad a \longmapsto \gamma \circ a.$$

The second condition can be pictured as



One may also allow Z to be an arbitrary scheme: for a $\mathbb{C}$-scheme S, morphisms $S \to Z$ correspond to $\mathbb{C}$-morphisms

$$S \longrightarrow Z \times_{\operatorname{Spec}\mathbb{Z}} \operatorname{Spec}\mathbb{C}.$$

Thus the formulation over $\mathbb{C}$ already gives this version of the universal property.

Remark 1.4.2 (Coarse does not mean fine)**.** Condition (I) classifies individual maps over $\mathbb{C}$. It does not assert that (1.4.1) is a bijection on families over every S. A *fine* moduli space represents the entire isomorphism-class functor and has a universal family. The presence of nontrivial automorphisms is the reason to retain this distinction.

References. For the stated universal property, see [FP97], §1.2, pp. 11–12. For the stack formulation and coarse moduli spaces, see [Alper26], §5.4, using the 2026 numbering.

1.5 Stability and Finiteness of Automorphisms

Definition 1.5.1 (Automorphism of a pointed map)**.** An automorphism of $(C;\{p_i\},\mu)$ is a scheme automorphism $\tau : C \to C$ with

$$\tau(p_i) = p_i \quad \text{for all } i, \qquad \mu \circ \tau = \mu.$$

We denote its group by $\operatorname{Aut}(C;\{p_i\},\mu)$.

Proposition 1.5.2 (The automorphism criterion)**.** *A pointed map from a quasi-stable curve is stable if and only if its automorphism group is finite.*

Proof. Let

$$G = \operatorname{Aut}(C;\{p_i\},\mu), \qquad \nu : \widetilde{C} = \coprod_{j=1}^{m} \widetilde{C}_j \longrightarrow C$$

be the normalization. Each $\widetilde{C}_j$ is a smooth irreducible projective curve. Put

$$g_j = g(\widetilde{C}_j), \qquad f_j = (\mu \circ \nu)|_{\widetilde{C}_j},$$

and define the finite set

$$Q_j = \widetilde{C}_j \cap \nu^{-1}(\{p_1,\dots,p_n\} \cup \operatorname{Sing} C), \qquad r_j = \#Q_j.$$

Thus Q_j includes the markings and *both branches* of every self-node on C_j.

The numerical condition on normalized components. Let δ_j be the number of self-nodes of C_j and let s_j count its special points as in Definition 1.1.5. Normalization gives

$$p_a(C_j) = g_j + \delta_j, \qquad r_j = s_j + 2\delta_j.$$

The genus formula follows from the normalization exact sequence; see [Har77], Chapter IV, Exercise 1.8. Consequently

$$2g_j - 2 + r_j = 2p_a(C_j) - 2 + s_j. \tag{1.5.1}$$

For a contracted component, the right side is positive precisely under the stability conditions: it is $-2 + s_j$ if $p_a(C_j) = 0$, is s_j if $p_a(C_j) = 1$, and is automatically positive for $p_a(C_j) \geq 2$. Thus stability is equivalent to

$$2g_j - 2 + r_j > 0 \qquad \text{for every contracted component } C_j. \tag{1.5.2}$$

For each normalized component define

$$A_j = \{\alpha \in \mathrm{Aut}(\widetilde{C}_j) : \alpha(q) = q \text{ for all } q \in Q_j, \quad f_j \circ \alpha = f_j\}.$$

Step 1: stability implies finiteness. Assume the pointed map is stable. If C_j is contracted, then f_j is constant and

$$A_j = \mathrm{Aut}(\widetilde{C}_j, Q_j).$$

By (1.5.2) this group is finite. We use here the classical finiteness theorem for automorphisms of a smooth pointed curve with $2g_j - 2 + r_j > 0$, stated in [Alper26], Proposition 6.1.8, p. 212.

If C_j is not contracted, let Y_j be its reduced integral image in X. The nonconstant map induces a finite extension of function fields

$$f_j^* : K_j := k(Y_j) \hookrightarrow k(\widetilde{C}_j).$$

Indeed, the extension is finitely generated and algebraic: both are function fields of integral curves, and the map is dominant onto its image. Every $\alpha \in A_j$ induces a field automorphism fixing K_j, because

$$\alpha^* \circ f_j^* = f_j^*.$$

An automorphism of a smooth integral curve is determined by its action on the function field, so

$$A_j \hookrightarrow \mathrm{Aut}_{K_j}\, k(\widetilde{C}_j), \qquad \#A_j \leq [k(\widetilde{C}_j) : K_j] < \infty.$$

There is no assertion of equality with the field-extension degree; such equality need not hold for a non-Galois extension. Hence A_j is finite for every j.

Let $Q = \coprod_j Q_j$ and $r = \#Q$. Every $\tau \in G$ lifts uniquely to an automorphism $\widetilde{\tau}$ of the normalization, with

$$\nu \circ \widetilde{\tau} = \tau \circ \nu.$$

The lift permutes the m components and the r points of Q. Thus we obtain a homomorphism

$$\rho : G \longrightarrow \mathfrak{S}_m \times \mathfrak{S}_r.$$

Let $G_0 = \ker \rho$. Its elements preserve each normalized component and fix all points of Q. Restriction gives

$$R : G_0 \longrightarrow \prod_{j=1}^{m} A_j, \qquad \tau \longmapsto (\widetilde{\tau}|_{\widetilde{C}_1}, \ldots, \widetilde{\tau}|_{\widetilde{C}_m}).$$

The restrictions belong to A_j, since

$$f_j \circ \widetilde{\tau}|_{\widetilde{C}_j} = \mu \circ \nu \circ \widetilde{\tau}|_{\widetilde{C}_j} = \mu \circ \tau \circ \nu|_{\widetilde{C}_j} = f_j.$$

The homomorphism R is injective. If $R(\tau)$ is the identity, then $\widetilde{\tau} = \mathrm{id}$ and $\tau \circ \nu = \nu$. Hence τ fixes every point of C. To conclude that it is the identity *as a scheme morphism*, use the injection

$$\mathcal{O}_C \hookrightarrow \nu_* \mathcal{O}_{\widetilde{C}}$$

for a reduced curve. Locally, with normalization map $\nu^{\#} : A \hookrightarrow B$, the diagram is

$$\begin{array}{ccc} A & \xhookrightarrow{\nu^{\#}} & B \\ {\scriptstyle \tau^{\#}}\uparrow & \nearrow_{\nu^{\#}} & \\ A & & \end{array}$$

and $\nu^{\#} \circ \tau^{\#} = \nu^{\#}$ implies $\tau^{\#} = \mathrm{id}_A$. Therefore $\tau = \mathrm{id}_C$.

We have proved

$$\#G_0 \le \prod_{j=1}^{m} \#A_j < \infty, \qquad \#G \le m!\, r! \prod_{j=1}^{m} \#A_j < \infty.$$

Step 2: finiteness implies stability. Assume now that G is finite. For a contracted component, take an element of

$$A_j = \mathrm{Aut}(\widetilde{C}_j, Q_j).$$

Extend it by the identity on all other normalized components. It fixes each preimage of every node, so it descends through the node identifications to an automorphism of C. It preserves every marking and commutes with μ, since f_j is constant. Thus $A_j \hookrightarrow G$, and A_j must be finite.

If (1.5.2) fails, there are four possibilities:

$$(g_j, r_j) \in \{(0,0), (0,1), (0,2), (1,0)\}.$$

Their pointed automorphism groups contain, respectively,

$$\mathrm{PGL}_2(\mathbb{C}), \quad \{z \mapsto z + b : b \in \mathbb{C}\}, \quad \{z \mapsto \lambda z : \lambda \in \mathbb{C}^*\},$$

and the translations of an elliptic curve. These groups are infinite, a contradiction. Therefore (1.5.2) holds for every contracted component, and the pointed map is stable. □

References. The statement is in [FP97], §1.2, p. 12. The proof above uses normalization as in [Alper26], Proposition 6.3.5, p. 226, while retaining the separate function-field argument for noncontracted components. Smooth pointed-curve finiteness is Proposition 6.1.8, p. 212.

1.5A The automorphism-free locus

For a projective target, the stable-map moduli stack is a separated Deligne–Mumford stack of finite type, with the above projective coarse moduli scheme. At a point with stabilizer G_x, the local coarse-space structure is described by an ètale chart of the form

$$[\operatorname{Spec} A/G_x] \longrightarrow \text{the moduli stack}, \qquad \operatorname{Spec} A^{G_x} \longrightarrow \overline{M}_{g,n}(X,\beta).$$

These charts form a cartesian square, with the lower map ètale. When $G_x = \{1\}$, the quotient chart is just $\operatorname{Spec} A$ and all its stabilizers are trivial. Since ètale morphisms are open, the locus of maps with no nontrivial automorphisms is open.

Definition 1.5.3. Write

$$\overline{M}^*_{g,n}(X,\beta) \subset \overline{M}_{g,n}(X,\beta)$$

for this open automorphism-free locus.

References. The local quotient statement is [Alper26], Corollary 5.4.13, *Local Structure of Coarse Moduli Spaces*, printed p. 190 in the January 5, 2026 draft.

1.6 Convex Targets and the Genus-Zero Boundary

Definition 1.6.1 (Convex variety)**.** A nonsingular variety X is *convex* if, for every morphism $\mu : \mathbb{P}^1 \to X$,

$$H^1(\mathbb{P}^1, \mu^* T_X) = 0.$$

Theorem 1.6.2 (Genus-zero maps to a convex target)**.** *Let X be a nonsingular projective convex variety.*

(i) The scheme $\overline{M}_{0,n}(X,\beta)$ is normal and projective. If nonempty, it is of pure dimension

$$\dim X + \int_\beta c_1(T_X) + n - 3.$$

(ii) It is locally a quotient of a nonsingular variety by a finite group.

(iii) The open locus $\overline{M}^*_{0,n}(X,\beta)$ is a nonsingular fine moduli space, equipped with a universal family of stable maps.

The word *normal* in part (i) does not assert irreducibility. The moduli scheme can be disconnected, reducible, or empty; the pure-dimension assertion concerns all of its components when it is nonempty. The distinction between the coarse space and the fine locus is retained in parts (ii) and (iii).

References. This is [FP97], §1.2, Theorem 2, p. 12. Its proof is in §§4–5 of that reference.

1.6A Reducible source curves

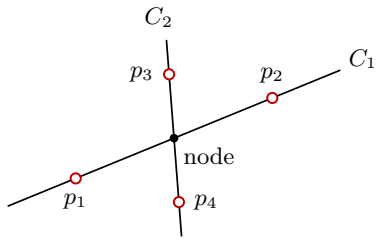
Definition 1.6.3 (Genus-zero boundary)**.** The *boundary* of $\overline{M}_{0,n}(X,\beta)$ is the locus of stable maps whose source curve is reducible.

A connected nodal curve of arithmetic genus zero is a tree of smooth rational components. In particular, its singular sources are precisely its reducible sources.

Example 1.6.4 (Two rational components)**.** A basic boundary source is

$$C = C_1 \cup C_2, \qquad C_1 \simeq C_2 \simeq \mathbb{P}^1,$$

with a single node. The following source has two markings on each component. If the target is a point, each component has three special points, including the node, and the resulting pointed curve is stable.



Theorem 1.6.5 (The boundary for a convex target)**.** *For a nonsingular projective convex variety X, the boundary of $\overline{M}_{0,n}(X,\beta)$ is a divisor with normal crossings, up to the finite group quotient of Theorem 1.6.2(ii).*

The statement means that on the nonsingular local covers, node-smoothing parameters give the boundary locally as

$$t_1 \cdots t_k = 0.$$

The finite stabilizer group can identify or permute branches before one passes to the coarse space. Thus the qualifier about finite quotients is part of the assertion.

References. See [FP97], §1.2, Theorem 3, p. 12, and the proof in §6.

1.6B The target is a point

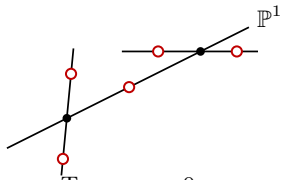
For $X = \operatorname{Spec}\mathbb{C}$ and $\beta = 0$, one obtains

$$\overline{M}_{0,n} = \overline{M}_{0,n}(\operatorname{Spec}\mathbb{C}, 0), \qquad n \geq 3.$$

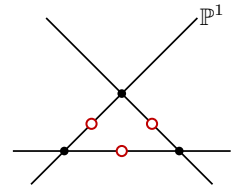
Its boundary $\overline{M}_{0,n} \setminus M_{0,n}$ is a divisor with *simple* normal crossings. The irreducible boundary divisors are smooth. See [Keel92], *Catalogue of results on the vital divisors*, Fact 1, printed p. 551; that fact refers to [Knudsen83], Theorem 2.7.

Example 1.6.6 (A tree and a cycle)**.** The left drawing below has three rational components arranged in a tree, with five markings. Each component has three special points. It is a stable genus-zero pointed curve and lies in the boundary of $\overline{M}_{0,5}$.

The triangular arrangement in the right drawing is also a stable pointed nodal curve: each rational component has two attachment nodes and one marking. Its arithmetic genus, however, is one. It is a genus-one contrast, rather than a point of the genus-zero boundary.



Tree: $p_a = 0$
a boundary point of $\overline{M}_{0,5}$



Cycle: $p_a = 1$
a point of $\overline{M}_{1,3}$

To check the genus, let Γ_C be the dual graph, with a vertex for each normalized component and an edge for each node. For a connected nodal curve,

$$p_a(C) = \sum_j g(\widetilde{C}_j) + b_1(\Gamma_C), \qquad b_1(\Gamma_C) = \#\text{edges} - \#\text{vertices} + 1.$$

The tree has two edges and three vertices, while the triangle has three edges and three vertices. Their arithmetic genera are therefore zero and one, respectively. This calculation also shows why a cycle of rational components cannot be used as a genus-zero source.

References. For the genus formula, see [Alper26], Proposition 6.2.12, printed p. 218, and for dual graphs, Definition 6.3.9, p. 227.

1.7 Evaluation Maps and Forgetting the Map

1.7A Independence of a projective embedding

Remark 1.7.1. The coarse moduli space does not depend on the projective embedding of X used in its construction. The functor of stable maps to X is intrinsic, and any two spaces with the coarse universal property are uniquely isomorphic: each universal property supplies a morphism to the other space, and uniqueness identifies both composites with the identity.

1.7B Evaluation as a natural transformation

The n markings induce canonical evaluation maps

$$\rho_i : \overline{M}_{g,n}(X,\beta) \longrightarrow X, \qquad 1 \le i \le n.$$

We construct them directly from the universal property, without assuming that the coarse space has a universal curve.

For each i, define

$$\theta_i : \overline{\mathcal{M}}_{g,n}(X,\beta) \longrightarrow h_X$$

by assigning to a family $\xi = (\pi : \mathcal{C} \to S, \{p_j\}, \mu)$ the map

$$\theta_i(S)([\xi]) = \mu \circ p_i \in \mathrm{Hom}_{\mathbb{C}}(S, X).$$

This is well defined on isomorphism classes: if $\tau : \mathcal{C} \to \mathcal{C}'$ is an isomorphism of pointed maps, then

$$\mu' \circ p_i' = \mu' \circ \tau \circ p_i = \mu \circ p_i.$$

It is natural under base change. For $a : T \to S$, pullback of the family gives

$$\theta_i(T)(a^*[\xi]) = (\mu \circ p_i) \circ a.$$

By Theorem 1.4.1, taking $Z = X$ and $\psi = \theta_i$ gives a unique morphism ρ_i with

$$\theta_i = \widetilde{\rho}_i \circ \varphi,$$

where

$$\widetilde{\rho}_i(S) : \mathrm{Hom}_{\mathbb{C}}(S, \overline{M}_{g,n}(X,\beta)) \longrightarrow \mathrm{Hom}_{\mathbb{C}}(S, X), \qquad a \longmapsto \rho_i \circ a.$$

Hence, for every family ξ over S,

$$\begin{aligned}\theta_i(S)([\xi]) &= \widetilde{\rho}_i(S)(\varphi_S([\xi])) \\ &= \rho_i \circ \varphi_S([\xi]),\end{aligned}$$

or, equivalently,

$$\mu \circ p_i = \rho_i \circ \varphi_S([\xi]) \quad \text{in } \operatorname{Hom}_{\mathbb{C}}(S, X). \tag{1.7.1}$$

For $S = \operatorname{Spec} \mathbb{C}$, this specializes to

$$\rho_i([C; \{p_j\}, \mu]) = \mu(p_i) \in X(\mathbb{C}).$$

Thus evaluation is a morphism on the coarse space even when the space is not fine.

1.7C Forgetting the map and stabilizing the source

Assume now

$$2g - 2 + n > 0. \tag{1.7.2}$$

After forgetting μ, the pointed domain need not be stable: a component that was non-contracted may have too few special points as a pointed curve. Stabilize the source by contracting the unstable rational components. Stabilization is canonical and compatible with base change, and yields a family of stable n-pointed genus-g curves.

This construction gives a natural transformation from the stable-map functor to the functor of stable pointed curves, and then to $h_{\overline{M}_{g,n}}$. Applying the coarse universal property with $Z = \overline{M}_{g,n}$ gives a morphism

$$\eta : \overline{M}_{g,n}(X, \beta) \longrightarrow \overline{M}_{g,n}.$$

On geometric points it sends a stable map to the stabilization of its marked source after the map to X is forgotten.

Condition (1.7.2) is necessary. For example, a degree-one map $\mathbb{P}^1 \to \mathbb{P}^1$ with no markings is a stable map, since its automorphism group over the target is trivial. But an unmarked genus-zero curve has no stable model. There is therefore no corresponding forgetful map to $\overline{M}_{0,0}$.

References. Evaluation and the forgetful map are in [FP97], §1.3, p. 13, where the inequality (1.7.2) is explicit. For stabilization of pointed curves and its functorial properties, see [Knudsen83], §2.

References

[FP97] W. Fulton and R. Pandharipande, *Notes on stable maps and quantum cohomology*, in *Algebraic Geometry—Santa Cruz 1995*, Proc. Sympos. Pure Math. **62**, Part 2, American Mathematical Society, 1997, 45–96. arXiv:alg-geom/9608011. Principal reference: §1, pp. 10–13 in the arXiv pagination. Theorems 1–3 are in §1.2; evaluation and forgetting are in §1.3.

[Alper26] J. Alper, *Stacks and Moduli*, draft dated January 5, 2026. Author's PDF. All numbers in this lecture refer to that version: Corollary 5.4.13, p. 190; Proposition 6.1.8, p. 212; Proposition 6.2.12, p. 218; Definition 6.2.18 and Examples 6.2.19–20, p. 220; Proposition 6.3.5, p. 226; Definition 6.3.9, p. 227; Proposition 6.3.22, p. 231. The linked draft may change; these references use the downloaded 2026 version.

[Har77] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics 52, Springer, 1977. doi:10.1007/978-1-4757-3849-0. Normalization and the genus formula: Chapter IV, Exercise 1.8. Finiteness of automorphisms of curves of genus at least two: Chapter IV, Exercise 5.2.

[Keel92] S. Keel, *Intersection theory of moduli space of stable n-pointed curves of genus zero*, Trans. Amer. Math. Soc. **330** (1992), 545–574. doi:10.2307/2153922. Smooth boundary divisors with normal crossings: *Catalogue of results on the vital divisors*, Fact 1, p. 551.

[Knudsen83] F. F. Knudsen, *The projectivity of the moduli space of stable curves, II: The stacks $M_{g,n}$*, Math. Scand. **52** (1983), 161–199. doi:10.7146/math.scand.a-12001. Stabilization: §2; smoothness and normal-crossings boundary: Theorem 2.7.

[Fulton98] W. Fulton, *Intersection Theory*, 2nd ed., Ergebnisse der Mathematik (3), vol. 2, Springer, 1998. doi:10.1007/978-1-4612-1700-8. Cycles and proper pushforward: Chapter 1; cycle-class maps: §19.1.

[Stacks] The Stacks Project Authors, *The Stacks Project*, stacks.math.columbia.edu. Connected reduced fibres and pushforward of the structure sheaf: Tag 0E0L. Relative dualizing sheaves: Tag 0E6P and Tag 0E6R. Smoothness: Tag 01V4. Sections as regular immersions: Tag 067R. Ampleness by component degrees on a proper curve: Tag 0B5Y.